Microcontroller Based Systems & Robotics Hub

Promoting robotic design and entrepreneurship experiences among students and teachers

Lesson 5:
Introduction to Arduino

CONTENTS



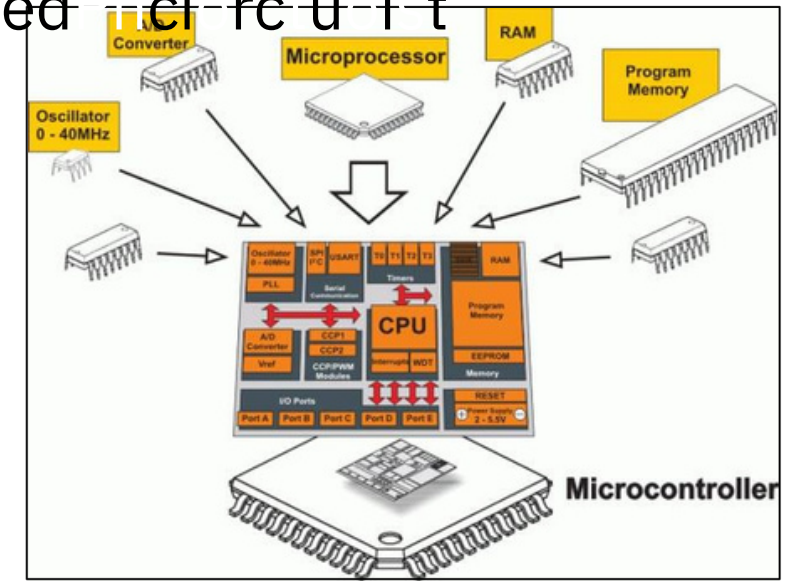
- Microcontrollers and microprocessors
- Introduction to Arduino
- Types of Arduino boards
- Programming basics: Structure of a code

- TASK/ACTIVITY:
  - Blink on-board LED
  - Blink LED(s)
  - Change brightness of LEDs

WHAT IS A MICROCONTROLLER?

- A compact integrated circuit on a single chip containing a processor, memory, and input/output as its main components
- Typically, it is "embedded" inside a
- device that is controlL E DS
  A microcontroller is often small and of low cost

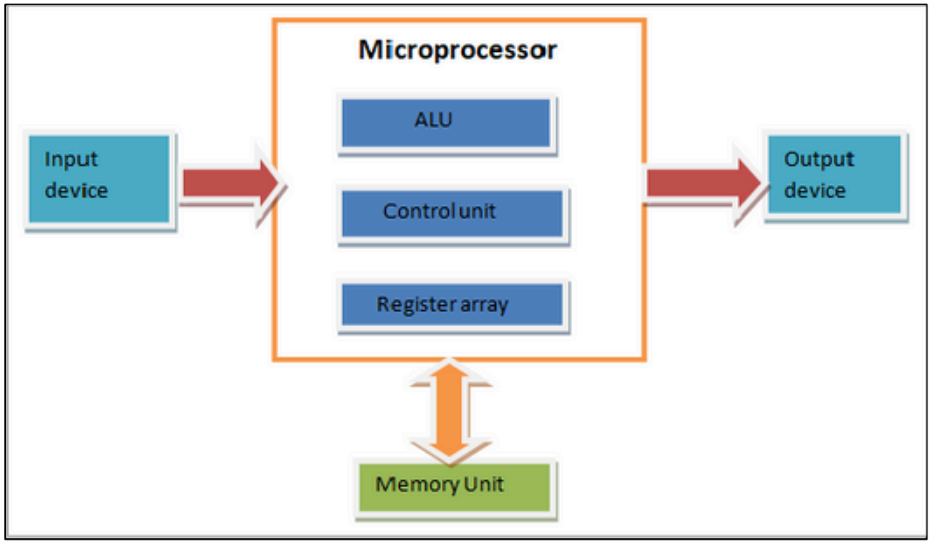




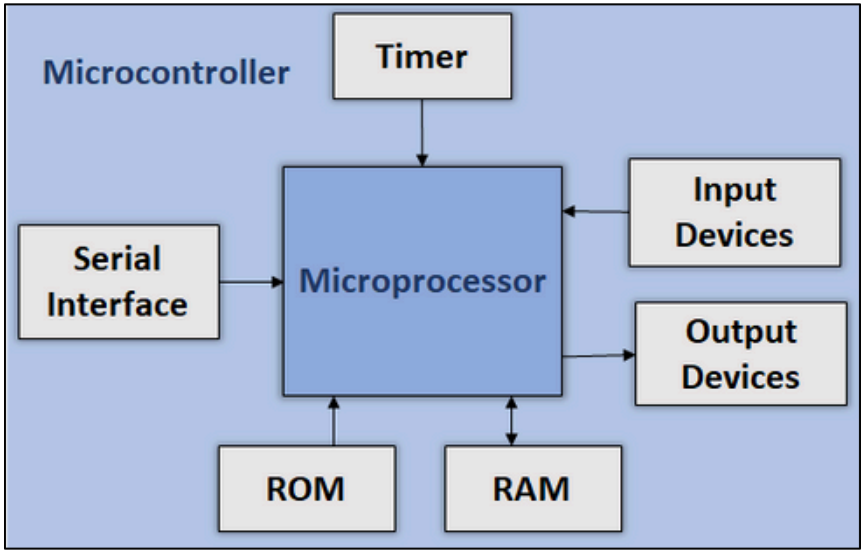
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MICROPROCESSOR VS MICROCONTROLLER

- The key difference: Microprocessor consists of only a Central Processing Unit, whereas the microcontroller contains a CPU, memory, I/O all integrated into one chip



Source



Source

MORE ABOUT THE DEVICES

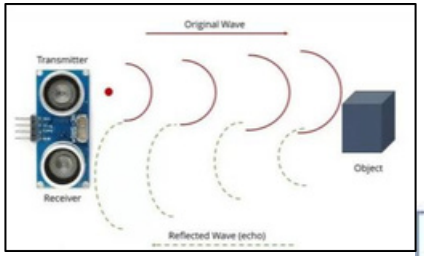
What is the difference between a computer, a microprocessor and a microcontroller ?

- Microprocessor is a full computation engine fabricated on a single chip; It acts as the central processing unit (CPU) of a microcomputer
  A Computer is a microprocessor
- many interfaces and memory chips packaged on a single circuit board with ; General purpose computers, i.e., PCs, are designed explicitly tointerface withhumans

- Microcontrollers are designed to interface, interact, andcommunicate with
➢ Electrical/electronic devices
➢ Sensor/actuators
➢ High-tech gadgets, etc.

COMPONENTS OF A ROBOT

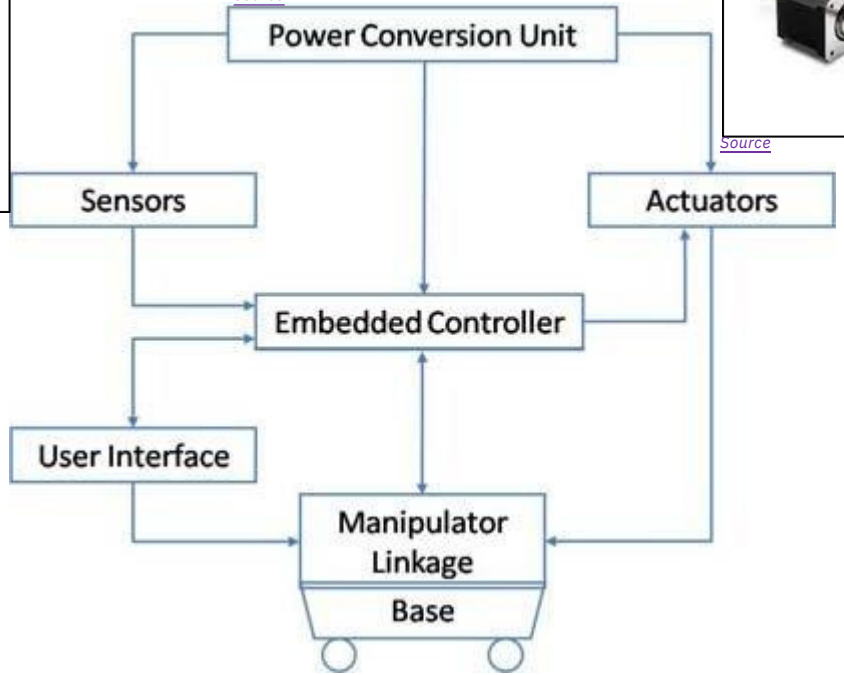
- As discussed earlier:



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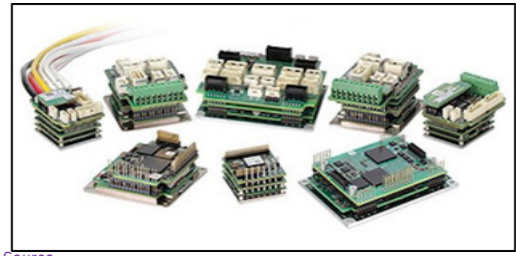


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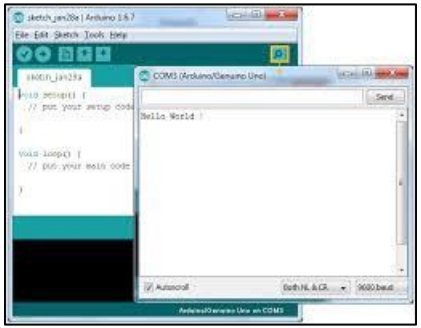




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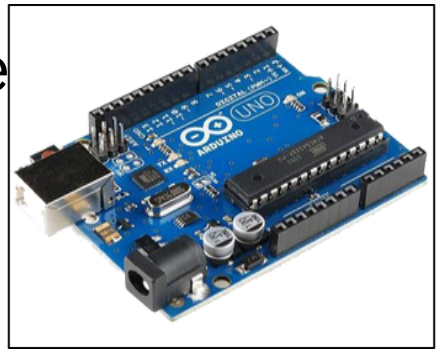
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Source

WHAT IS ARDUINO???

- Arduino is an open-source electronic platform for easy use of hardware and software
- It can sense the environment by receiving input from variety of sensors and make decision and then give the output
- It can be used to develop stand-alone interactive objects or can be connected to software on your computer



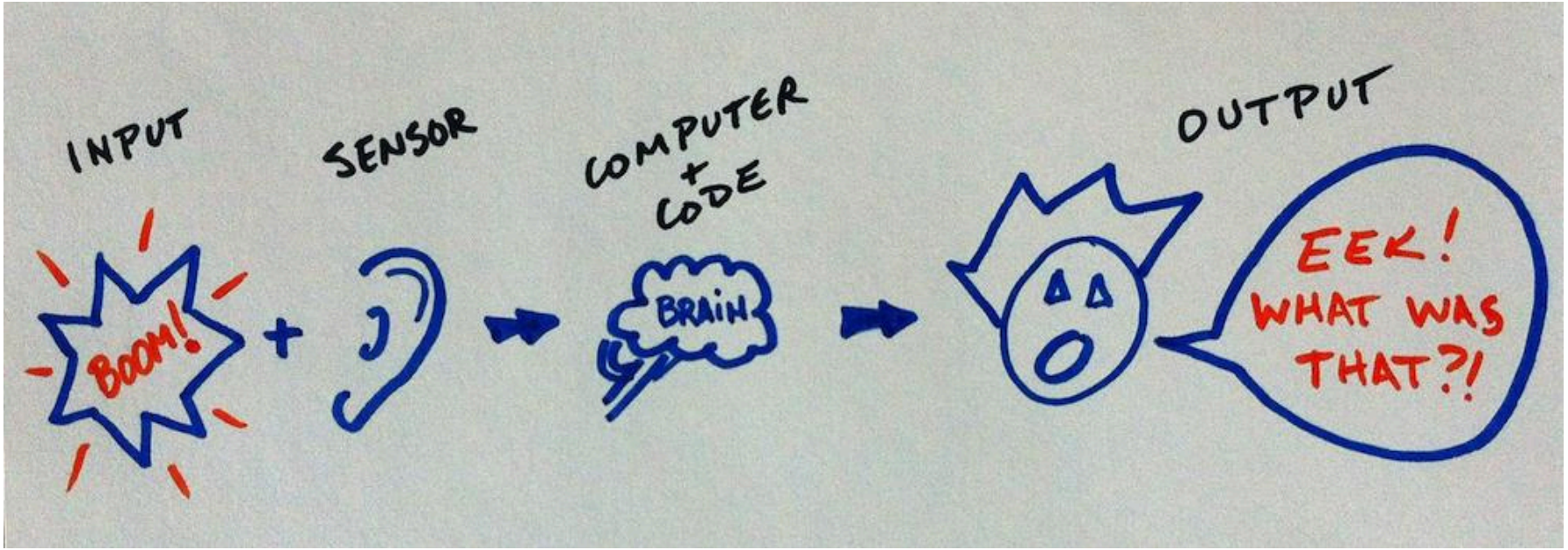
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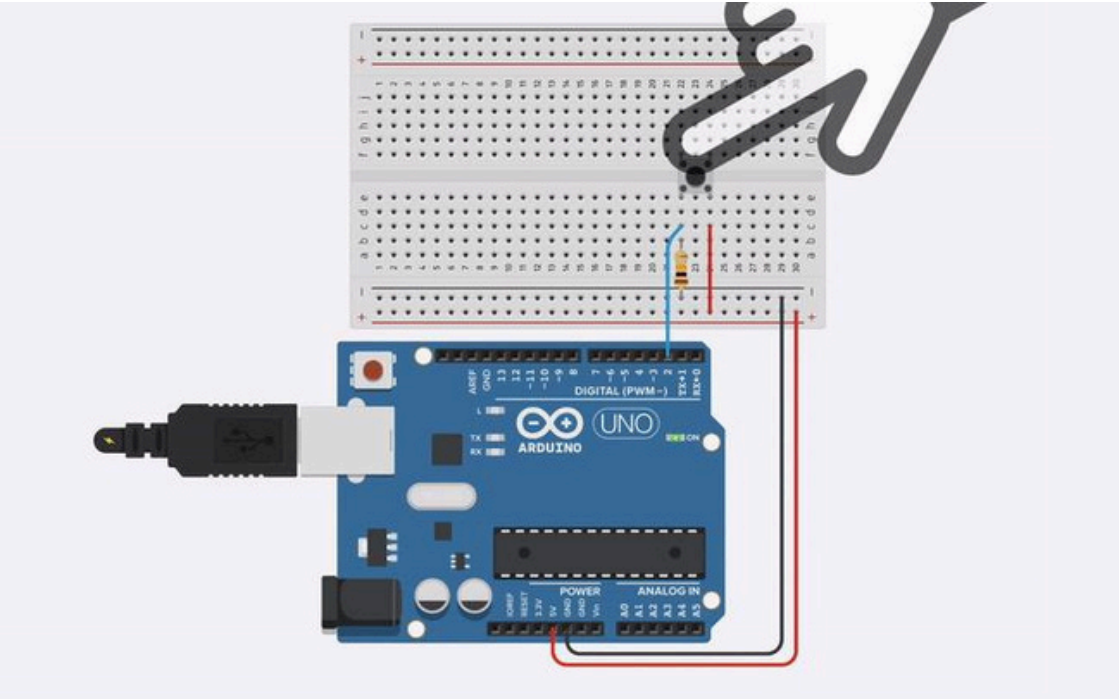
Arduino

WHAT DOES IT DO???

- Working: The microcontroller (computer) is programmed (code) to receive information (input) from the sensors and the outputis given through computer (IDE) or other peripherals



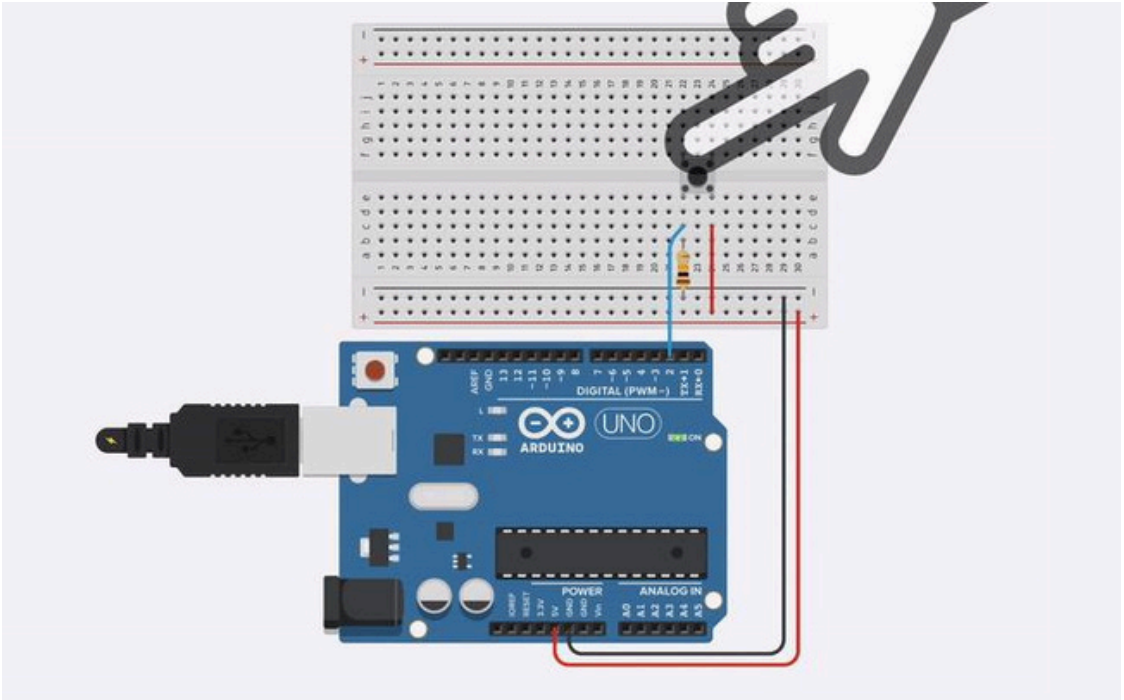
ACTIVITY 1



Source

- What is the circuit doing?
- What elements do you see on the circuit?
- Is there any input?
- If yes, what is the input?
- Is there any output?
- If yes, what is the output?
-

ACTIVITY 1 -SOLUTION



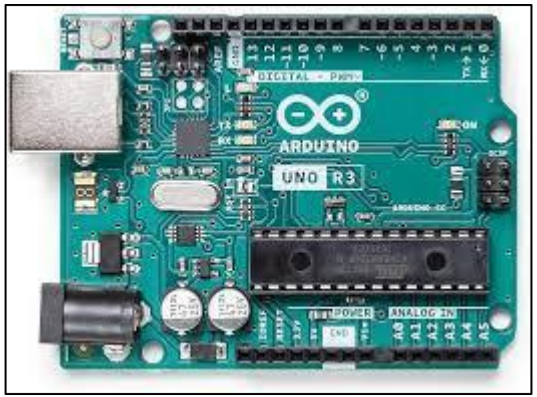
Source

- What is the circuit doing?
  Light up the on-board LED on button press
- What elements do you see on the circuit?
  Button, resistor, wires
- Is there any input? Yes
- If yes, what is the input?
  Button
  Is there any output?
- If yes, what is the output?
  Yes
- LED

ARDUINO

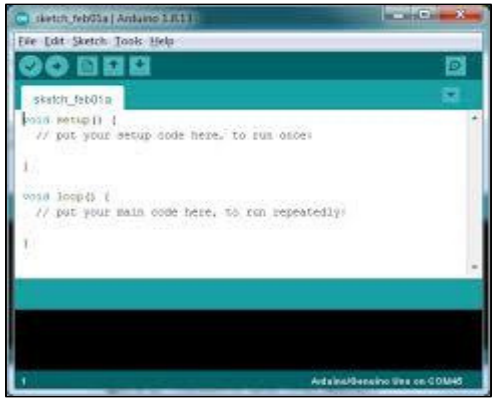
- The word Arduino can mean 3 things:

Hardw are



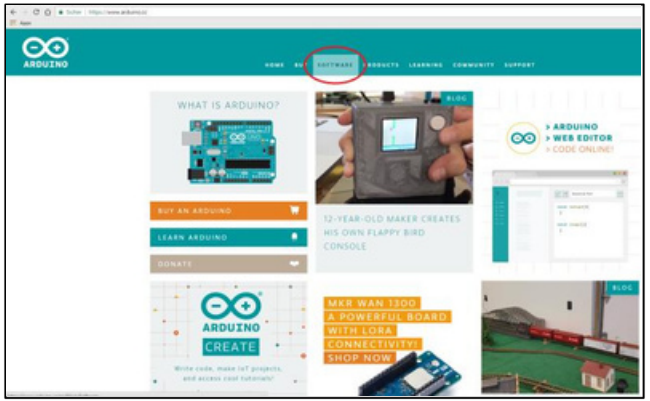
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An interface



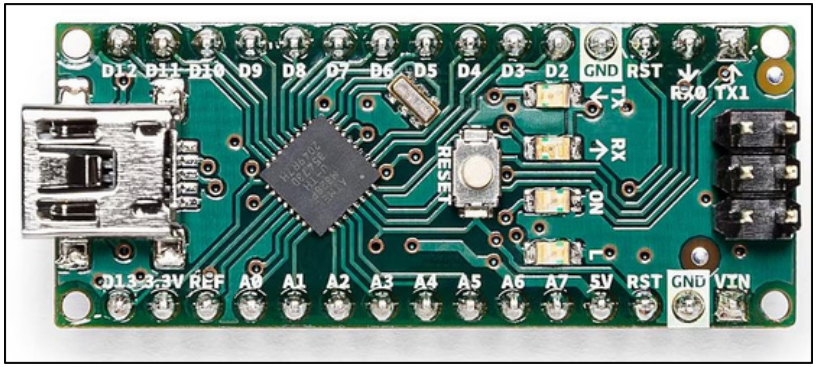
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A community

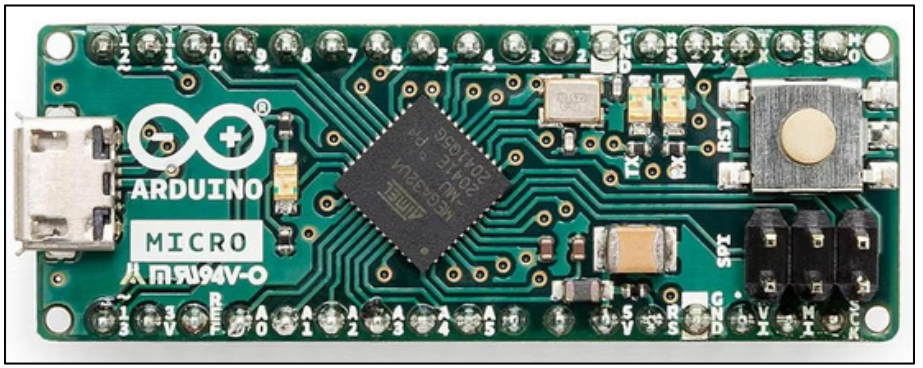


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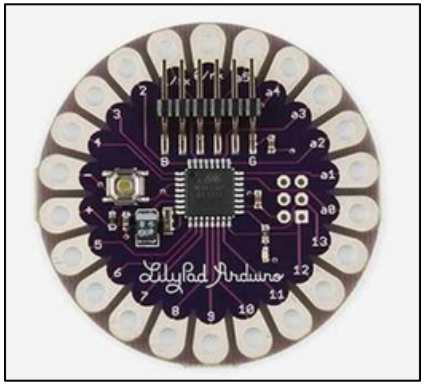
DIFFERENT TYPES OF ARDUINO



Arduino NANO

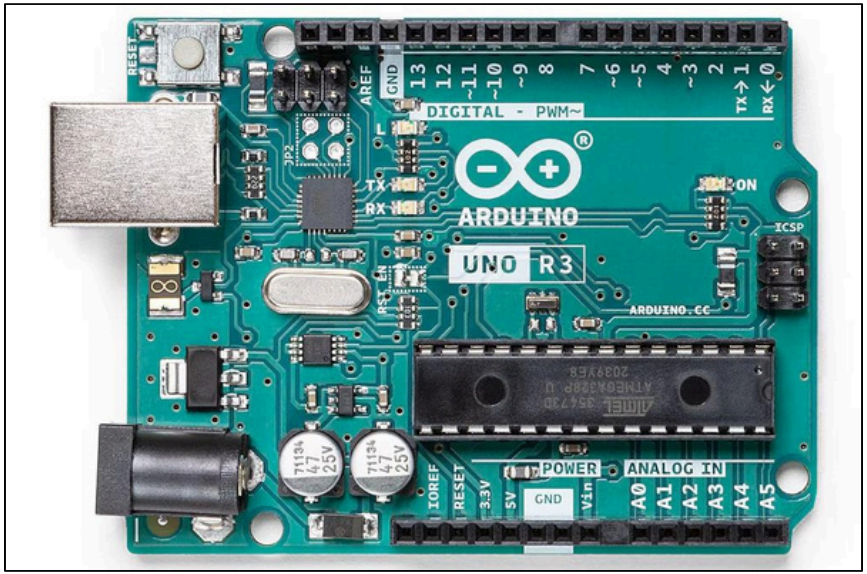


Arduino Micro

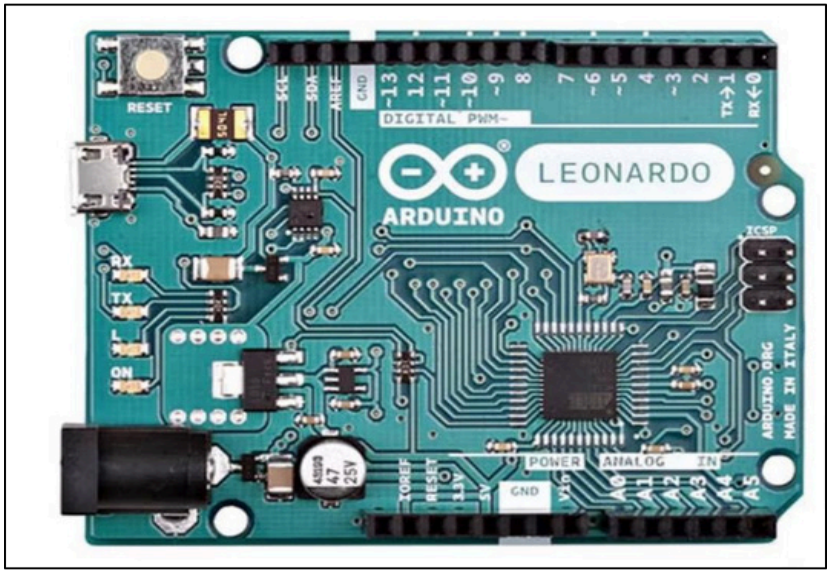


Arduino LilyPad

DIFFERENT TYPES OF ARDUINO

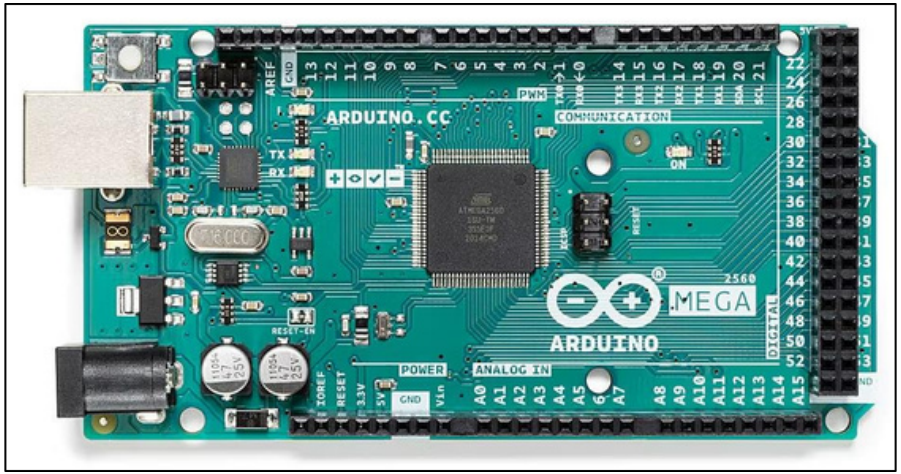


Arduino UNO R3

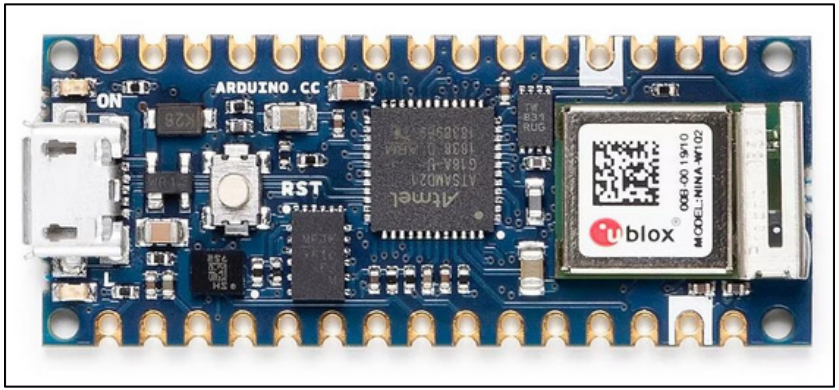


Arduino Leonardo

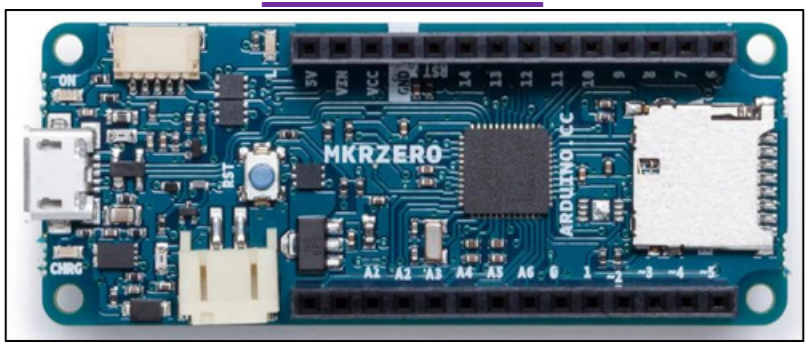
DIFFERENT TYPES OF ARDUINO



Arduino MEGA 2560 R3

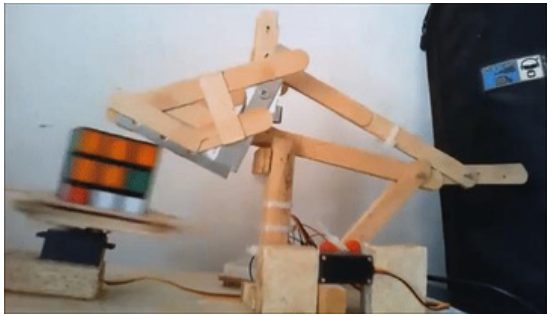


Arduino IOT board

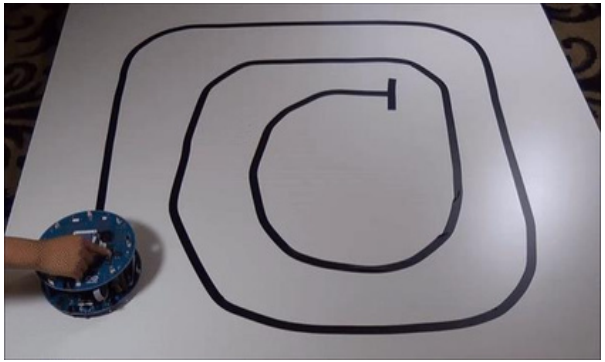


Arduino MKR ZERO

PROJECTS BASED ON ARDUINO



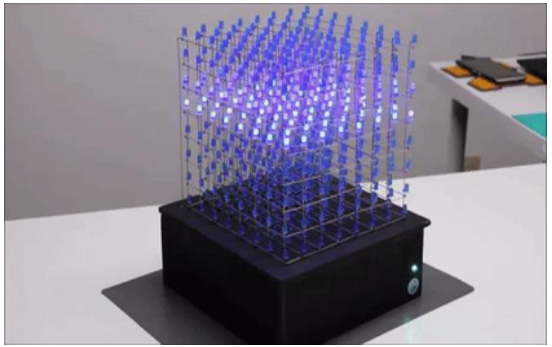
Rubik's cube solver



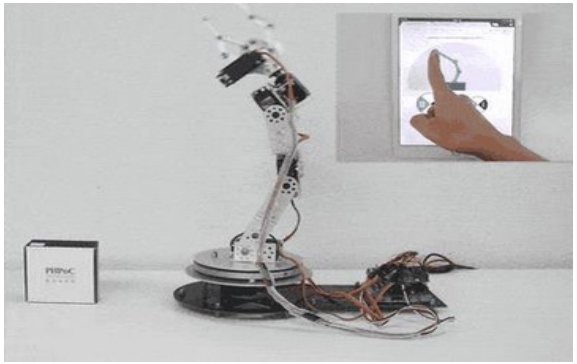
Line following Turtlebot



Spider bot with Arduino



LED cube powered by Arduino

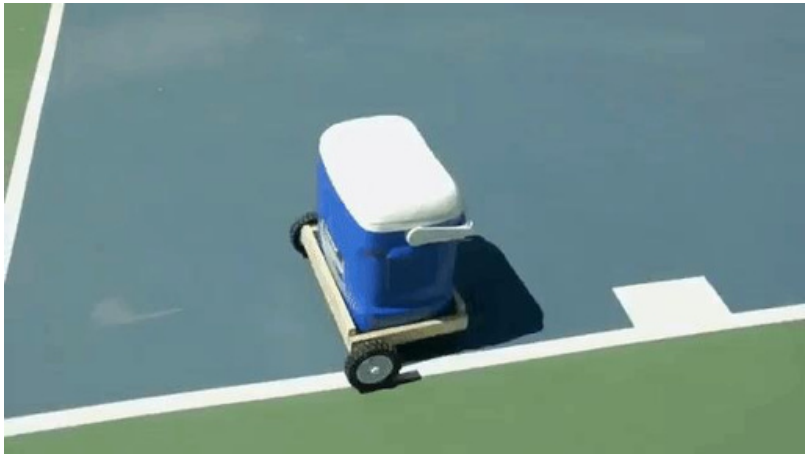


App controlled trainable arm with Arduino

PROJECTS BASED ON ARDUINO

- An example of a human-following cooler called, "Follow me"





Follow me cooler with Arduino

PROJECTS BASED ON ARDUINO -NYU

- Human-following luggage with Arduino:

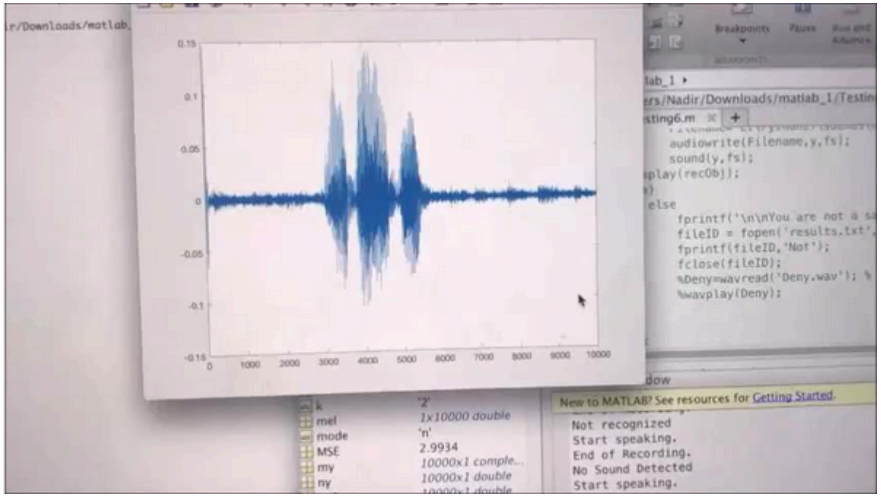


PROJECTS BASED ON ARDUINO

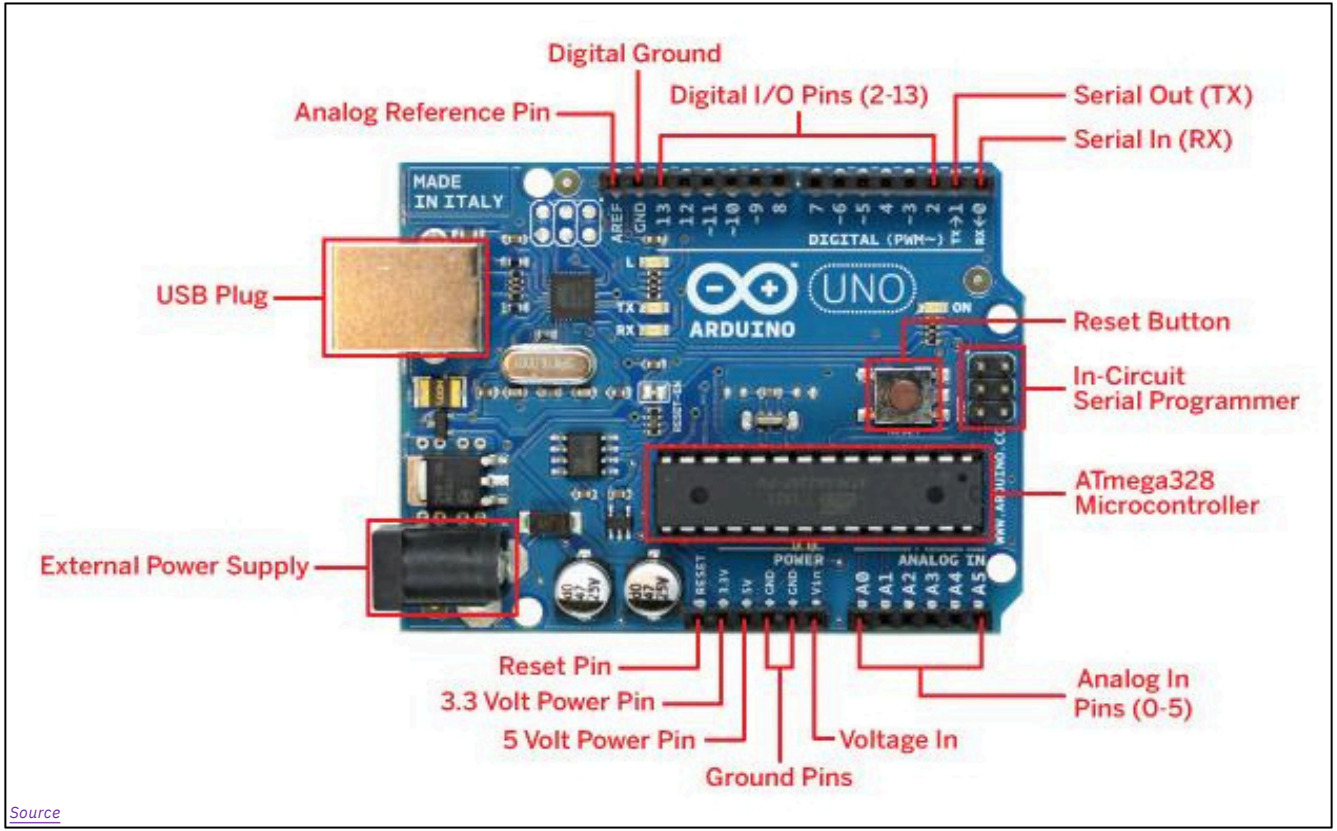
- Jarvis –Voice assistant with Arduino:



Source



ARDUINO UNO DEVELOPMENT BOARD



Source

ARDUINO HARDWARE SUMMARY

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	16 MHz

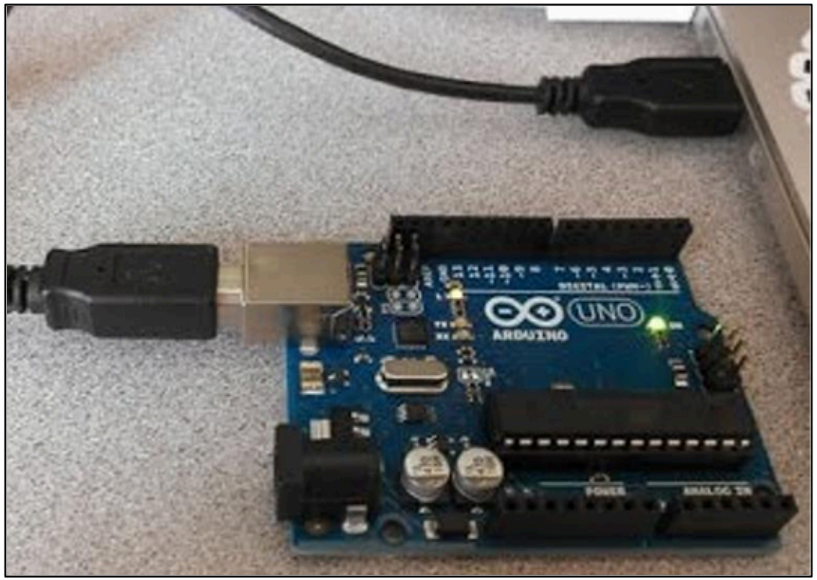
ACTIVITY 2

Check out: http://arduino.cc/en/Guide/HomePage

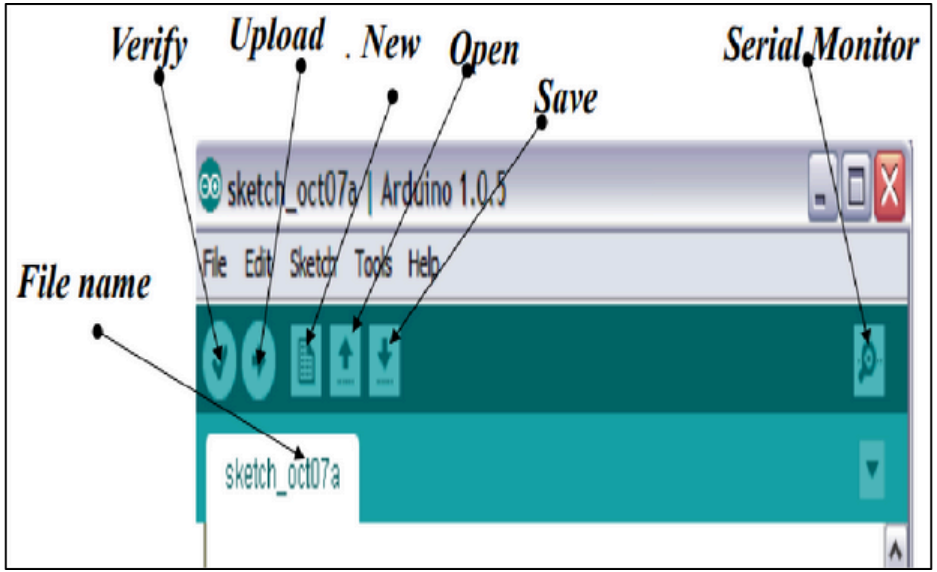
1. Download and install the Arduino environment (IDE) according to your system (Mac or Windows)
2. Connect the board to your computer via the USB cable
3. If needed, install the drivers
4. Launch the Arduino IDE

.POWERING UP THE ARDUINO WITH USB

- Connecting via USB cable to load the code and provide pow er at the same time

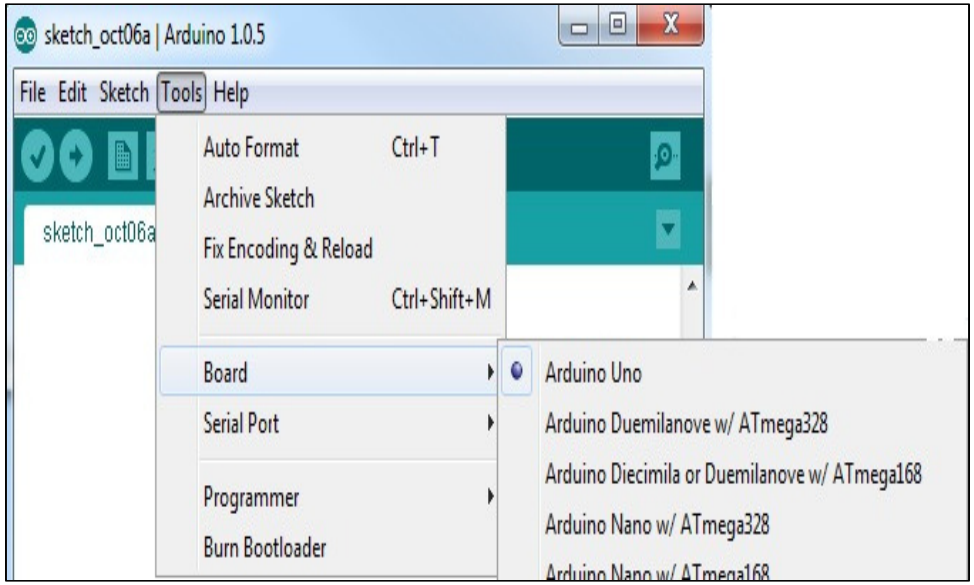


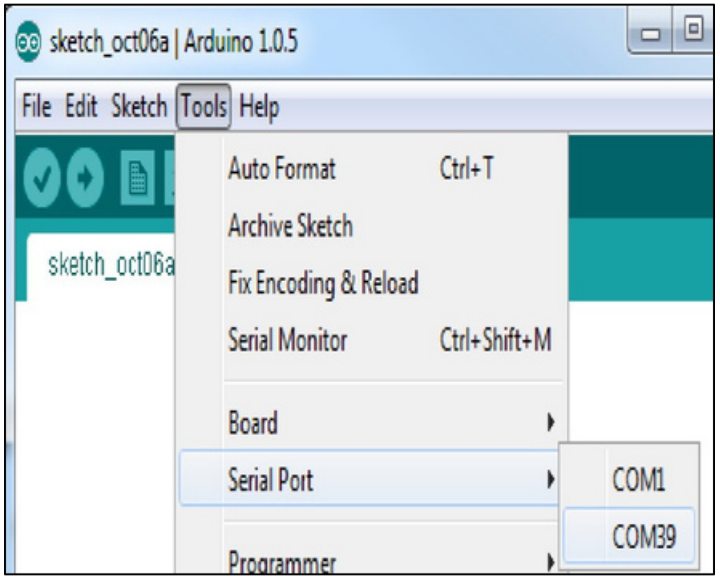
Source



SELECT SERIAL PORT AND BOARD

- Select the board Arduino UNO and the port showing in the Serial Port section





ACTIVITY 3



Source

Program to blink the on-board LED

1. Select your board
2. Selectyour serialport
3. Open the blinkexample
4. Uploadthe program

ACTIVITY 3 -SOLUTION

- Go to File >Examples > Digital > Blink

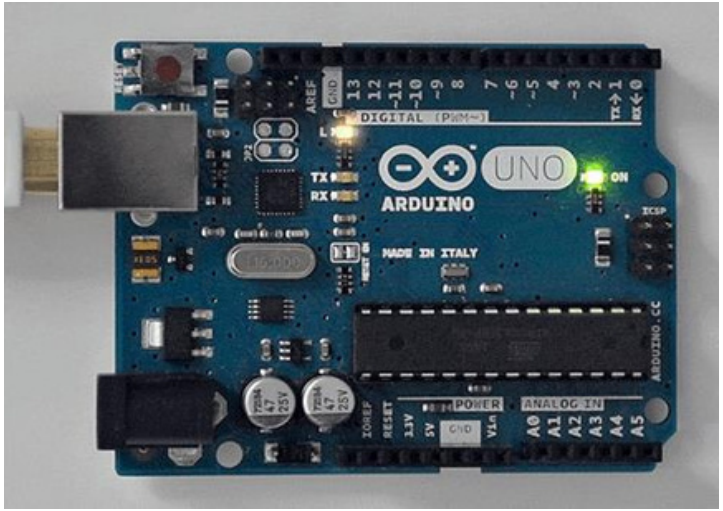
Blink §

```
/*
  Blink

  https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink
*/

// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);   // turn the LED on (HIGH is the voltage level)
  delay(1000);                       // wait for a second
  digitalWrite(LED_BUILTIN, LOW);    // turn the LED off by making the voltage LOW
  delay(1000);                       // wait for a second
}
```



Source

A LITTLE BIT ABOUT THE CODE

Sketch: Arduino code is referred to as sketch and consists of:

void setup() {

- Instructions between the two curly brackets will be run only once when the Arduino program first runs and used for the purposeofsetup
- Initialize I/O pins (directions), initialize serial communication, etc.

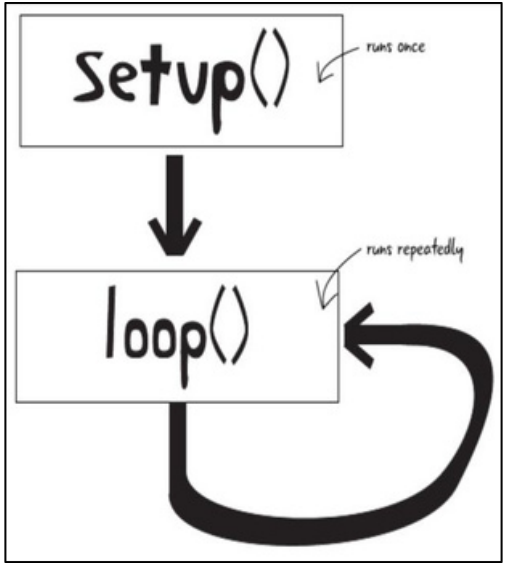
}

void loop() {

- Thisfunction is run after setup has finished
- Itruns in a continuousloop until power is removed

}

Sketch



Source

BASIC SYNTAX FOR ARDUINO PROGRAMS

Basic rules of programming Arduino :

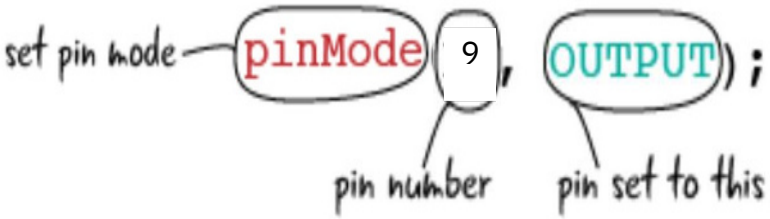
- // Single line comment
- /*
  Multi line comment, or block-comment
- */
- {} Curly braces to indicate starting and ending of a block of code unbalanced curly braces will cause compile-time errors
- ; Each line of code must end with a semicolon
  NOTE: Semicolon should not be used after #include, #define and for “if, else statements, for, while loops” if code is inside the curly braces

PROGRAMMING: I/O

pinMode(pin, mode)

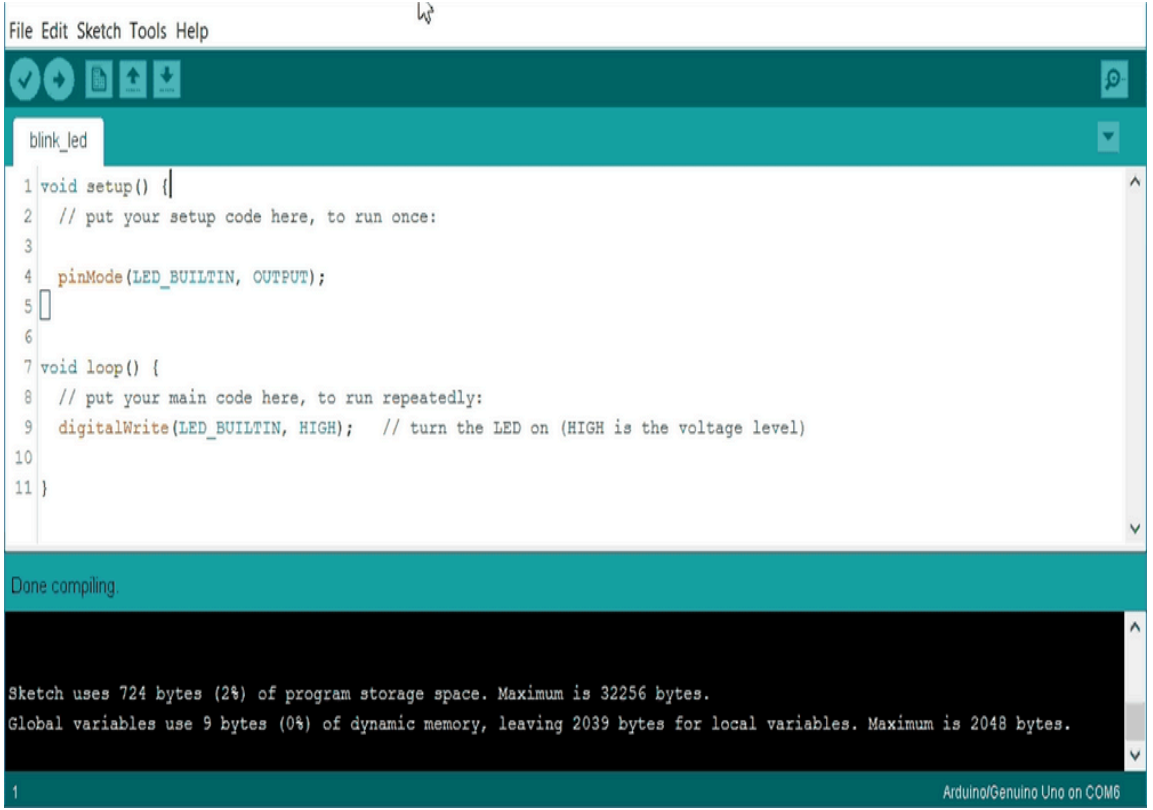
- pin refers to digital I/O pin number and mode argument refers to INPUT or OUTPUT
- Can usedigital I/Opin 2to13 (0and1used for RX andTX)
- Digital I/O pins have default mode as INPUT, no need to explicitly declare as inputs

➢ Connection: Connect an OUTPUT pin to an external device in series with a 470Ω or 1KΩ resistor



UPLOADING YOUR CODE

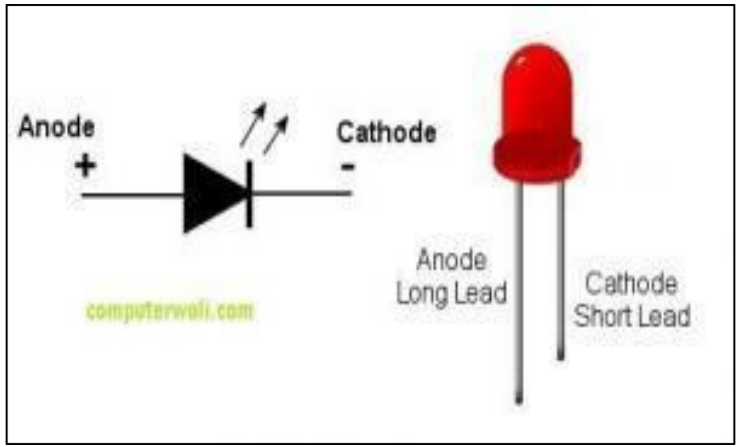
- Select your board Arduino UNO
- Select your port from tools
- Upload the code to the Arduino as shown
- Check for errors
- Check if the code is compiled successfully



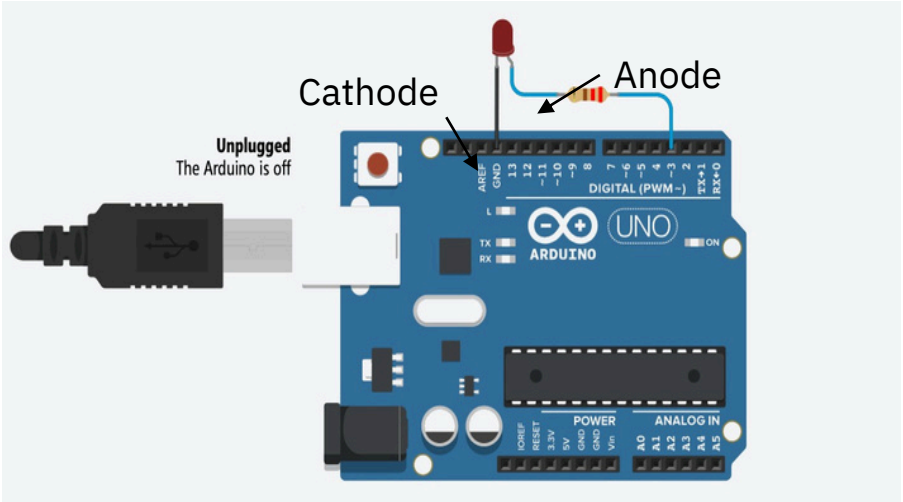
BLINK AN EXTERNAL LED

Things to remember:

1. Longer leg (Anode) of the LEDgoes to one end of the resistor in series and the other end of resistor goes to pin (3)here
2. Shorter leg (Cathode) goes to ground (GND)



Source



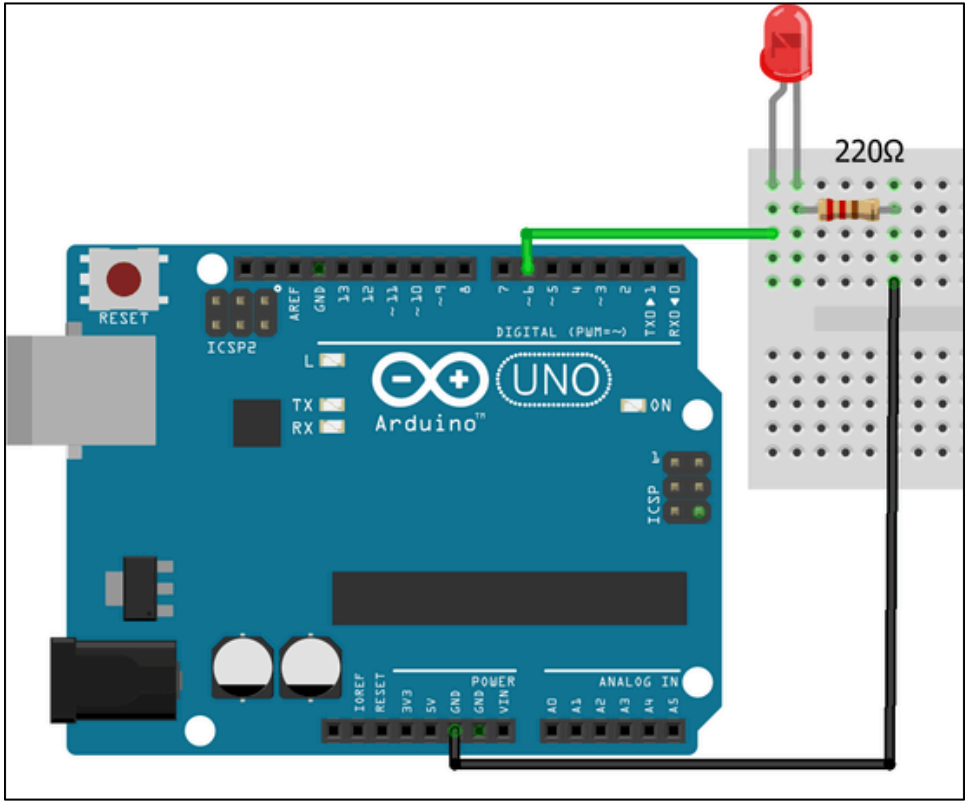
Blink LED

ACTIVITY 4

- Write a program to blink one external LED using Arduino

ACTIVITY 4 -SOLUTION

- Connections:
  Pins: 6, GND



Source

ACTIVITY 4 -SOLUTION

- Code:

Blink_oneExternalLED | Arduino 1.8.19

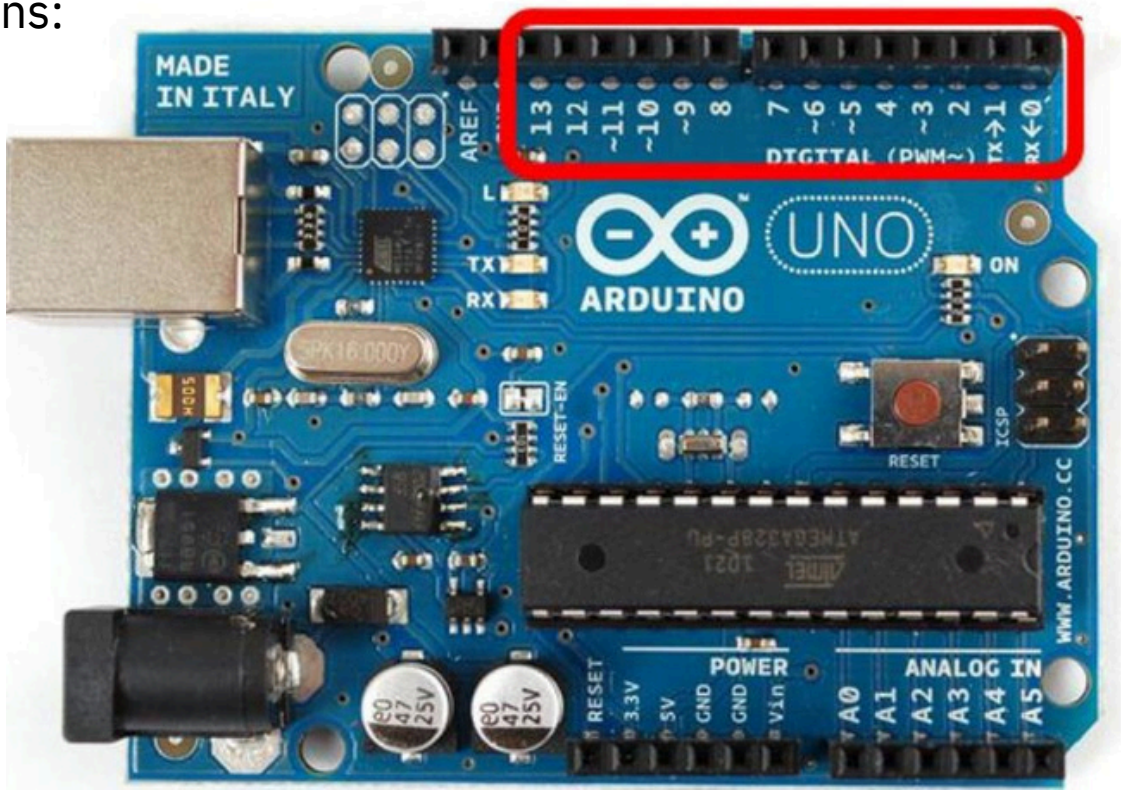
File Edit Sketch Tools Help

Blink_oneExternalLED §

```
void setup() {
pinMode(6, OUTPUT);  // initialize digital pin 6 as an output
}
void loop() {
digitalWrite(6, HIGH);  // turn the LED on (HIGH is the voltage level)
delay(1000);     // wait for a second
digitalWrite(6, LOW);   // turn the LED off by making the voltage LOW
delay(1000);             // wait for a second
}
```

PROGRAMMING ESSENTIALS: I/O

- Digital I/O pins:



PROGRAMMING ESSENTIALS: I/O

digitalRead(pin)

- Reads the value from a specified digital pin with the result being 1 (HIGH) or 0 (LOW)

• Pincan bespecified as avariable or constant (0-13)

➢ Syntax: value = digitalRead(pin); //sets 'value' equal to the state of the input pin

digitalWrite(pin, value)

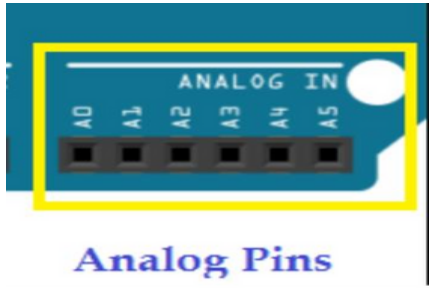
• Pinrefers toadigital pin, canbeavariable or aconstant(0-13)
• Value is either logic level HIGH/LOW, TRUE/FALSE, 1/0
• Outputs either logic level HIGH or LOW at a specified digital pin

➢ Syntax: digitalWrite(pin, value); //sets 'value' as the state of the input pin

PROGRAMMING ESSENTIALS: I/O

analogRead(pin)

- Reads the valuefrom a specifiedanalog pinwith a 10-bit resolution
- Pinargument can be specifiedas a variableor constant (0-5)
- Resulting integer values range from 0 to 1023 and must be scaled to appropriate units



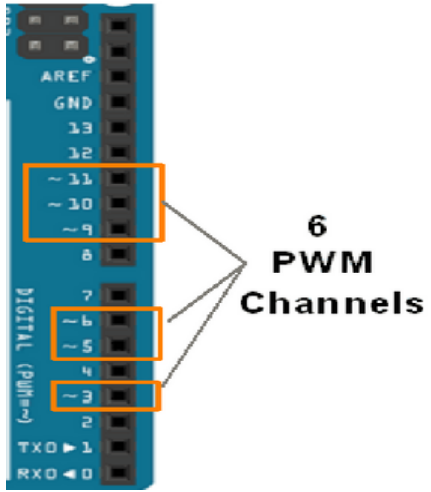
Analog Pins

➢ Syntax: value = analogRead(pin); //sets 'value' equal to analog reading on 'pin'

analogWrite(pin, value)

- Outputs a PWM signal to the specifiedoutput pin
- This function works on pins3, 5, 6, 9, 10, and 11
- Value can be specifiedas a variableor constant with range 0-255

➢ Syntax: analogWrite(pin, value); //writes 'value' to 'pin'



ACTIVITY 5

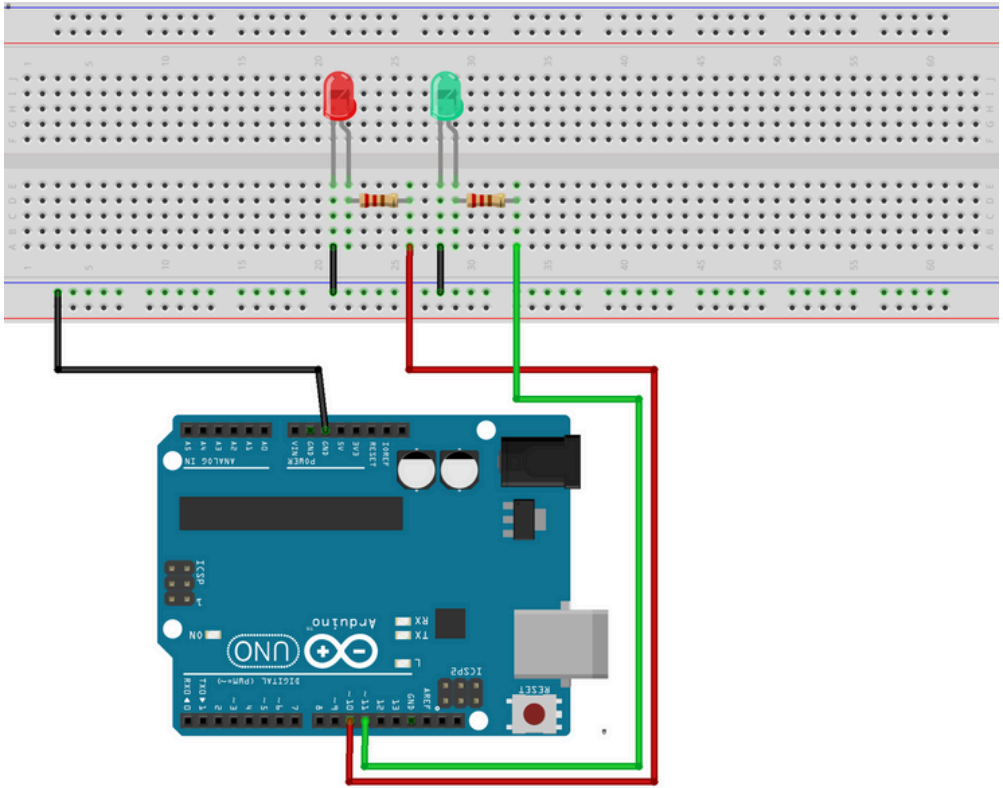
- Attach 2 LEDs to your Arduino, each to a separate pin, blink one LED only once and the other repeatedly



Source

ACTIVITY 5 -SOLUTION

- Connections:
  Pins: 10, 11, GND



Source

Source

ACTIVITY 5 -SOLUTION

- Code:

Blink_twoExternalLEDs | Arduino 1.8.19

File Edit Sketch Tools Help

Blink_twoExternalLEDs §

```
void setup()
{
pinMode(11,OUTPUT);
pinMode(10,OUTPUT);
delay(100);

// Blink LED on pin 11 once
digitalWrite(11, HIGH);
delay(1000);
digitalWrite(11, LOW);
delay(1000);
}
```

```
void loop()
{
// Blink LED on pin 10 forever
digitalWrite(10, HIGH);
delay(1000);
digitalWrite(10, LOW);
delay(1000);
}
```

Serial.begin(rate)

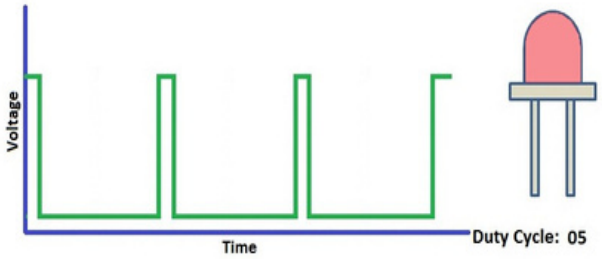
- Opensserial port and sets thebaud rateforserial data transmission
- Typical baud rate for communicating with PC is 9600 although other speeds are supported
- When using serial communication, digital pins 0 (RX) and 1 (TX) cannot be used at thesametime

delay(ms)

- Pauses the program for time as specified in milliseconds, with 1000 being equal to 1 second

PWM BASIC CONCEPTS

Pulse width modulation (PWM): Converts a digital value to analog output
Pulse width modulation (PWM) allows Arduino to generate a series of pulses
When a pin is output high, the apparent voltage at that pin will be close to 5 V
When the pin is made output low it is close to 0 V



Source



Source

PWM BASIC CONCEPTS

- Duty cycle: Higher the duty cycle, higher is the voltage
- Duty specifies the analog output level as a fraction of 256ths of 5 V ranging from 0 to 4.98 V

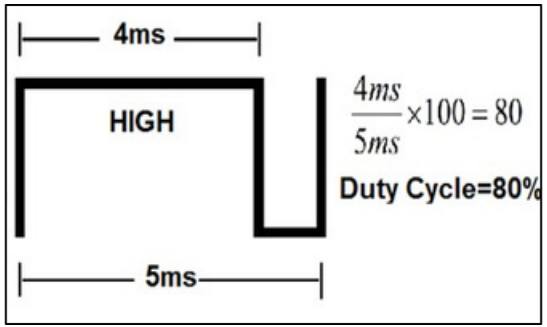
$$\text{duty cycle} = \frac{t_{ON}}{t_{ON} + t_{OFF}}$$

t_{ON} = ON time

t_{OFF} = OFF time

$t_{ON} + t_{OFF}$ = Time period

Source



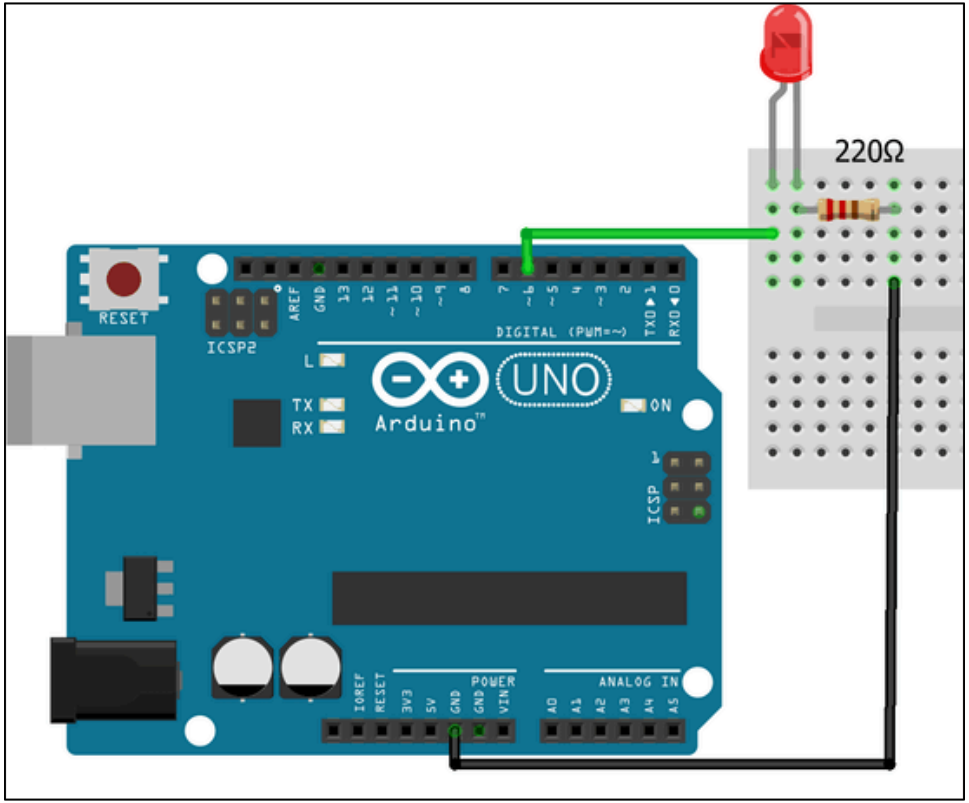
Source

ACTIVITY 6

- Attach one LED to your Arduino and write a program to change its brightness

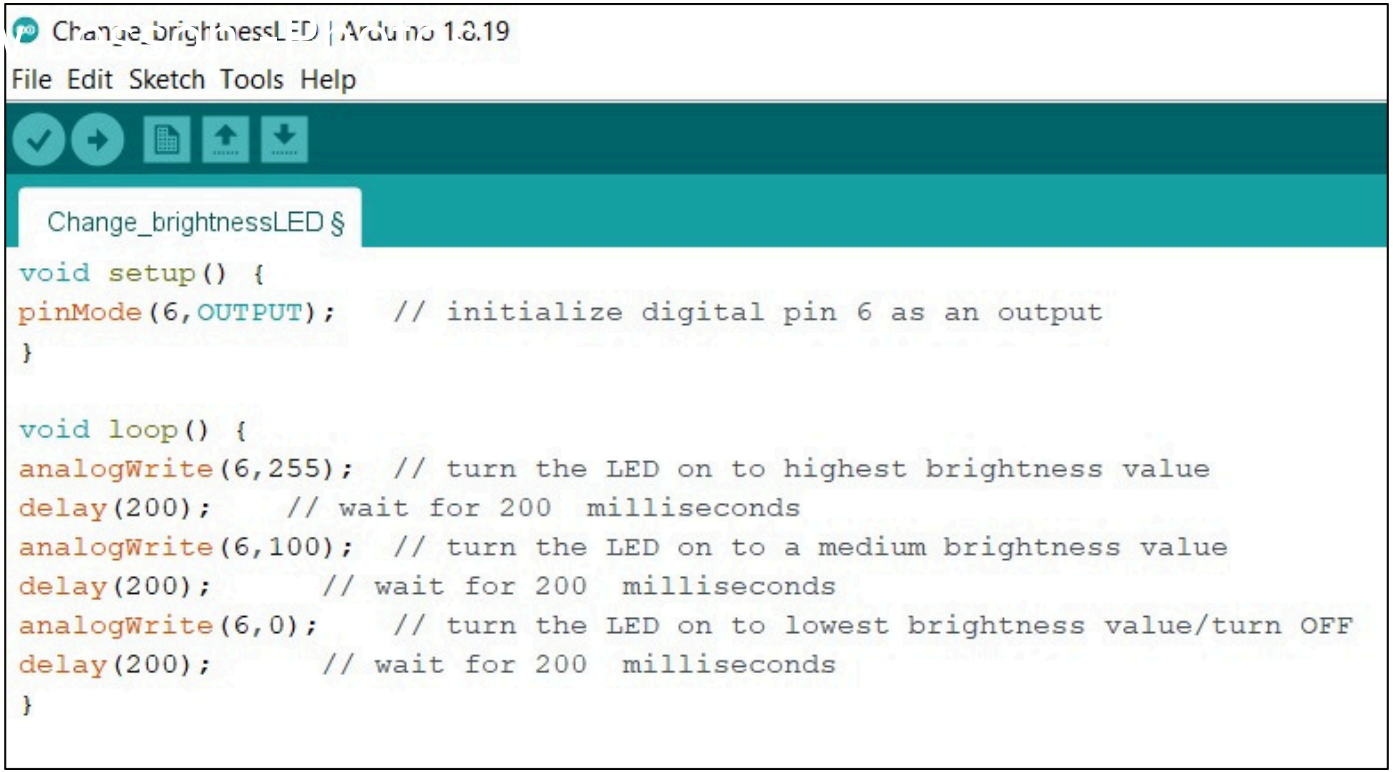
ACTIVITY 6 -SOLUTION

- Connections:
  Pins: 6, GND



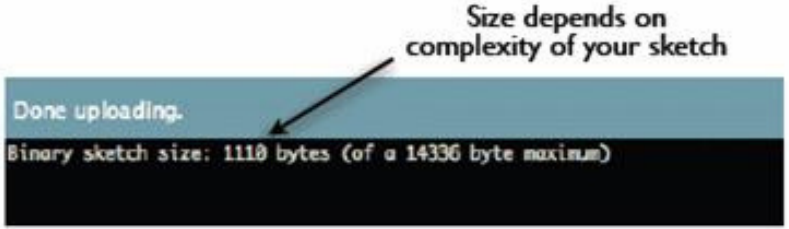
ACTIVITY 6 -SOLUTION

- Code:



STATUS MESSAGES

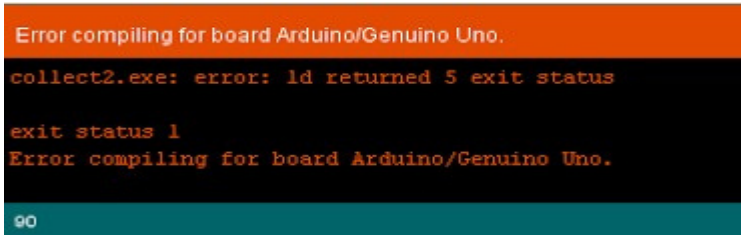
Successful upload:



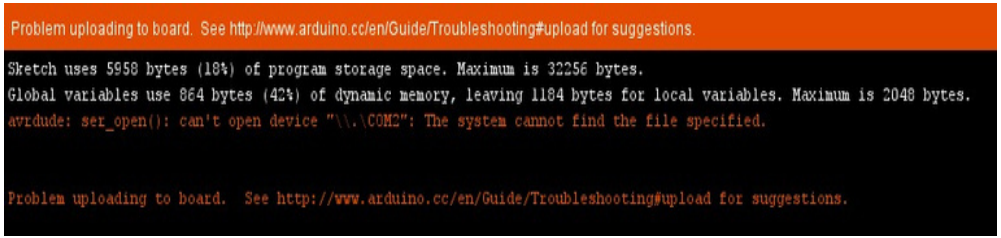
Wrong port selected:



Error in compiling:



Wrong board selected:

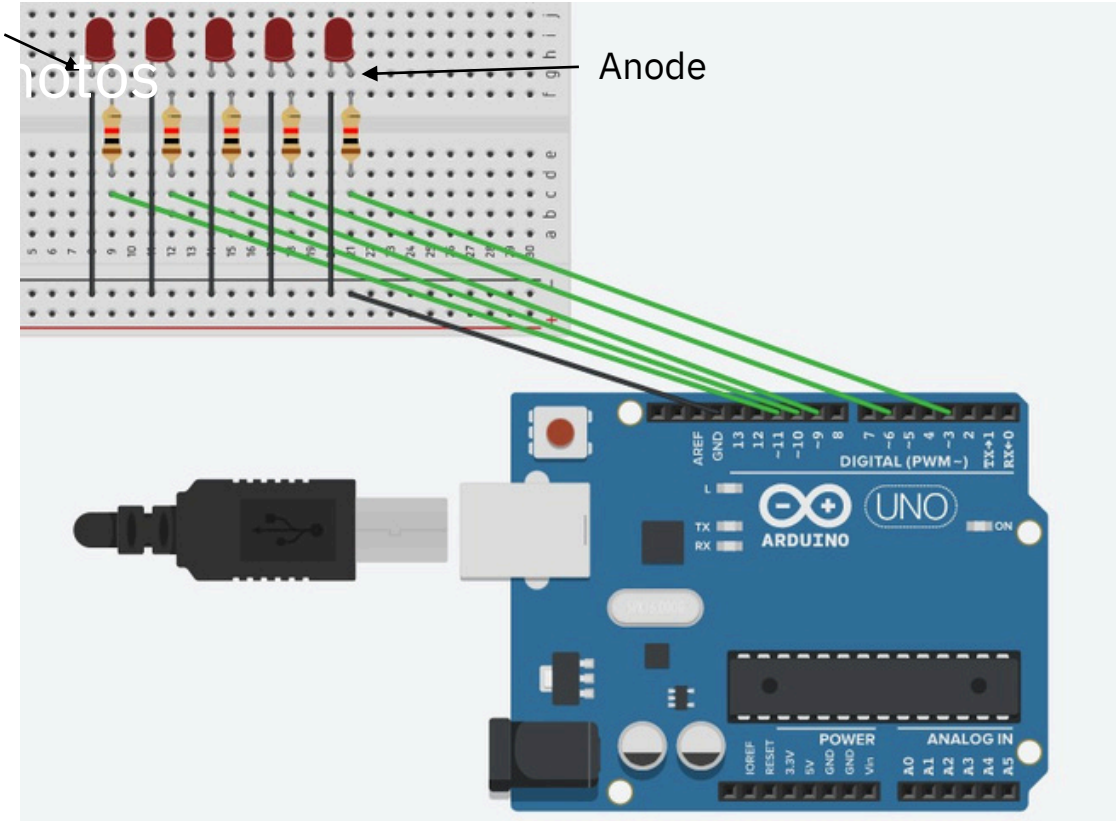


ACTIVITY 7

- Attach at least 5 LEDs to your Arduino, each to a separate pin. Blink them together

ACTIVITY 7 -SOLUTION

- Connections:
  Pins: 3, 6, 9, 10, 11, GND



ACTIVITY 7 -SOLUTION

- Code:

Blink_5LEDs | Arduino 1.8.19

File Edit Sketch Tools Help

Blink_5LEDs

```
void setup()
{
pinMode(11,OUTPUT);   // initialize digital pin 11 as an output
pinMode(10,OUTPUT);   // initialize digital pin 10 as an output
pinMode(9,OUTPUT);    // initialize digital pin 9 as an output
pinMode(6,OUTPUT);    // initialize digital pin 6 as an output
pinMode(3,OUTPUT);    // initialize digital pin 3 as an output
}
```

ACTIVITY 7 -SOLUTION

```
void loop()
{
// make pins 3, 6, 9, 10, 11 HIGH

digitalWrite(11, HIGH);
digitalWrite(10, HIGH);
digitalWrite(9, HIGH);
digitalWrite(6, HIGH);
digitalWrite(3, HIGH);

// Delay for a second
delay(1000);
```

```
// make pins 3, 6, 9, 10, 11 LOW

digitalWrite(11, LOW);
digitalWrite(10, LOW);
digitalWrite(9, LOW);
digitalWrite(6, LOW);
digitalWrite(3, LOW);

// Delay for a second
delay(1000);
}
```

VARIABLES

- A variable is a place to store a piece of data
- It has a name, a value, a type and size accordingly

- Syntax: variable_type variable_name = value;

Type	Bytes(size)	Example
boolean	1	boolean led_on=True;
char	1	char char_1 = ‘a’;
int	2	int temp = 48;
float	4	float height = 2.5;
long	4	long time = 5;

VARIABLES

- Save the results in different variable types:

int a = 50;
int b = 30;
int c = 0;
c = a / b;
Output: 1
	int a = 50;
	int b = 30;
	float c = 0;
	c = a / b;
	Output: 1.00
		float a = 50;
		float b = 30;
		float c = 0;
		c = a / b;
		Output: 1.66
			float a = 50;
			float b = 30;
			int c = 0;
			c = a / b;
			Output: 1

ACTIVITY 8

- Divide 5 by 2 and store the value as int and as float

ACTIVITY 8 -SOLUTION

int a = 5;
int b = 2;
int c = 0;
c = a / b;
 	
Output : 2
	int a = 5;
	int b = 2;
	float c = 0;
	c = a / b;
	
	Output : 2.00



Thank You!

Questions and Feedback?